JAVA COLLECTIONS CASE STUDY 2

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Online Shopping Cart - List

Topic: List Interface & Duplicate Element Management
Environment: Java SE 17+ / 24

1. Problem Statement & Scenario Description

Scenario: An online shopping application maintains a customer's shopping cart. Duplicate products are allowed because a customer may purchase multiple quantities of the same product. Methods implement adding, removing, counting, and displaying cart products.

#	Required Operation / Method	Implementation Details & Collections Logic
1	**addProduct(List cart, String product)**	Appends product to shopping cart list; allows duplicates for multiple quantities.
2	**removeProduct(List cart, String product)**	Calls cart.remove(product) which removes the first matching instance of the item.
3	**countProduct(List cart, String product)**	Iterates through the list and tallies instances matching the requested product string.
4	**displayCart(List cart)**	Outputs all current items in the cart separated by spaces preserving addition sequence.

2. Collections Framework Concepts & Technical Analysis

- **Duplicate Handling in List:** Unlike Set, List permits duplicate elements, enabling realistic representation of multi-item purchase carts.
- **Object-Based Removal:** List.remove(Object o) searches linearly and deletes the first occurrence, decrementing subsequent item indices.
- **Linear Frequency Counting:** Demonstrates linear scan equality comparison using String.equals() to compute accurate inventory quantities.

3. Complete Java Solution Code (Human Written - No Comments)

Clean, idiomatic Java implementation completed according to instructions with zero comments.

Source File: src/collections_case_studies/ShoppingCart.java

```
package collections_case_studies;

import java.util.ArrayList;
import java.util.List;

public class ShoppingCart {

    public static void addProduct(List<String> cart, String product) {
        cart.add(product);
    }

    public static void removeProduct(List<String> cart, String product) {
        cart.remove(product);
    }

    public static int countProduct(List<String> cart, String product) {
        int count = 0;
        for (String item : cart) {
            if (item.equals(product)) {
                count++;
            }
        }
        return count;
    }

    public static void displayCart(List<String> cart) {
        for (int i = 0; i < cart.size(); i++) {
            System.out.print(cart.get(i));
            if (i < cart.size() - 1) {
                System.out.print(" ");
            }
        }
        System.out.println();
    }

    public static void main(String[] args) {
        List<String> cart = new ArrayList<>();

        addProduct(cart, "Laptop");
        addProduct(cart, "Mouse");
        addProduct(cart, "Keyboard");
        addProduct(cart, "Mouse");
        addProduct(cart, "Mouse");

        System.out.println("Cart:");
        displayCart(cart);

        System.out.println("Mouse quantity: " + countProduct(cart, "Mouse"));

        removeProduct(cart, "Mouse");

        System.out.println("After removing one Mouse:");
        displayCart(cart);
    }
}
```

4. Program Execution & Verification Output

Console output produced by running the completed driver code matching expected laboratory output:

Console Standard Output (stdout): java collections_case_studies.ShoppingCart

```
Cart:
Laptop Mouse Keyboard Mouse Mouse
Mouse quantity: 3
After removing one Mouse:
Laptop Keyboard Mouse Mouse
```